

Rules of Inference

Logical inference

- A type of logical argument

- Given a collection of *premises*, deduce a *conclusion*

"There is blood on Zora's knife. Whoever's knife has blood on it is the killer. Therefore, Zora is the killer"

- An inference is *valid* if whenever each of the premises is true the conclusion is also true

"If pigs fly, then I can fly. Pigs fly. Therefore I can fly."

- *Rules of inference* give small, *valid* patterns of inference which we can use to build larger *arguments*

Example

Consider the following arguments:

*“If you have a current password, then you can log onto the network.
You have a current password. Therefore, you can log onto the
network.”*

and

*“If you have a current password, then you can log onto the network.
You can log onto the network. Therefore, you have a current
password.”*

Inference in symbolic form

● Write these arguments in symbolic form

p - you have a current password

q - you can log onto the network

$$\frac{p \rightarrow q}{p}$$

$\therefore q$

$$\frac{p \rightarrow q}{q}$$

$\therefore p$

‘Therefore’

“If you have a current password,
then you can log onto the network.
You have a current password.
Therefore, you can log onto the
network.”

“If you have a current password,
then you can log onto the network.
You can log onto the network.
Therefore, you have a current
password.”

Inference and Tautologies

- If both $p \rightarrow q$ and p are true, since the implication $p \rightarrow q$ is true and p is true, q must also be true.
- Equivalently, $\Phi = ((p \rightarrow q) \wedge p) \rightarrow q$ is a tautology:

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	Φ
0	0	1	0	1
0	1	1	0	1
1	0	0	0	1
1	1	1	1	1

The first example is a **valid** argument!

Inference and Tautologies

● Let us try $\Psi = ((p \rightarrow q) \wedge q) \rightarrow p$

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge q$	Ψ
0	0	1	0	1
0	1	1	1	0
1	0	0	0	1
1	1	1	1	1

In the case $p = 0$, $q = 1$ both conditions $(p \rightarrow q)$ and q are true, but p is false. This is not a valid argument! It is a **logical fallacy**

General Definition of Inference

- The general form of an inference in symbolic logic is

$$(p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n) \rightarrow q$$

premises

conclusion

- The inference is **valid** if whenever each of the premises is true the conclusion is also true
- Equivalently, the inference is valid if and only if the above formula is a tautology

Rules of Inference

- *Checking that a complicated inference is valid is generally hard, even for computer.*
- *Instead, build complicated valid arguments by breaking down into known, valid **rules of inference**.*

● *E.g., Modus ponens*

$$\begin{array}{c} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

*“If you have a current password, then you can log onto the network.
You have a current password.
Therefore, you can log onto the network.”*

Rule of Syllogism



$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$



The corresponding tautology $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

*“If you send me an e-mail, then I’ll finish writing the program.
If I finish writing the program, then I’ll go to sleep early.”*

p - ‘you will send me an e-mail’

q - ‘I will finish writing the program’

r - ‘I will go to sleep early’

“Therefore, if you send me an e-mail, then I’ll go to sleep early”

Modus Tollens



$$p \rightarrow q$$

$$\neg q$$

$$\therefore \neg p$$



The corresponding tautology $((p \rightarrow q) \wedge \neg q) \rightarrow \neg p$

“If today is Friday, then tomorrow I’ll go skiing”.

“I will not go skiing tomorrow”.

p - ‘today is Friday’

q - ‘I will go skiing tomorrow’

“Therefore, today is not Friday”

Rule of Disjunctive Syllogism



$$\underline{p \vee q}$$

$$\neg p$$

$$\therefore q$$



The corresponding tautology $((p \vee q) \wedge \neg p) \rightarrow q$

“I’ll go skiing this weekend.

I will not go skiing on Saturday.”

p - ‘I will go skiing on Saturday’

q - ‘I will go skiing on Sunday’

“Therefore, I will go skiing on Sunday”

Rule for Proof by Cases



$$\begin{array}{l} p \rightarrow r \\ q \rightarrow r \\ \hline \therefore (p \vee q) \rightarrow r \end{array}$$



The corresponding tautology $((p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow ((p \vee q) \rightarrow r)$

“If today is Saturday, then I’ll go skiing.

If today is Sunday, then I’ll go skiing.

p - ‘today is Saturday’

q - ‘today is Sunday’

r - ‘I’ll go skiing’

“Therefore, if today is Saturday or Sunday, then I will go skiing”

Rules of Contradiction, Simplification, and Addition

● *Rule of Contradiction* *Reductio ad Absurdum*

$$\frac{p \rightarrow F}{\therefore \neg p}$$
 The corresponding tautology $(p \rightarrow F) \rightarrow \neg p$

● *Rule of Simplification*

$$\frac{p \wedge q}{\therefore p}$$
 The corresponding tautology $(p \wedge q) \rightarrow p$

● *Rule of Addition*

$$\frac{p}{\therefore p \vee q}$$
 The corresponding tautology $p \rightarrow (p \vee q)$

Fallacies (invalid inferences)

● *Affirming the conclusion*

$$\begin{array}{c} p \rightarrow q \\ q \\ \hline \therefore p \end{array}$$

The corresponding proposition
 $((p \rightarrow q) \wedge q) \rightarrow p$

● *Denying the hypothesis*

$$\begin{array}{c} p \rightarrow q \\ \neg p \\ \hline \therefore \neg q \end{array}$$

The corresponding proposition
 $((p \rightarrow q) \wedge \neg p) \rightarrow \neg q$

Practice

Exercises from the Book:

7th edition: No. 3, 5, 6, 11, 12 (page 78)

8th edition: No. 3, 5, 6, 11, 12 (page 82)